D-Flow / QGF
Overnight investigation

Third campaign | November 6-7, 2026
LAYOUT DRAFT - RESULTS STILL IN PROGRESS

Heldout evidence cutoff: 2026-11-07T05:40:24.742443+00:00

The current comparison uses four D-Flow families: regular or stochastic, each with raw or nearby-action averaged Q. Each family has a separate λ0 and λ1 condition, giving eight D-Flow conditions. All use identity-Jacobian guidance and complete eight-slice flow endpoints.

The controls are BC, original QGF at fixed κ=.02/.04, independently tuned original QGF, one original IPG and one original FK corrector. The target is 40 heldout episodes per unique selected configuration and trained model, with training seeds 11, 23 and 37.

Closed heldout evidence in this snapshot	Episode rows
New physical episodes, unique CSV batches	200
Inherited complete A20 reference episodes	200
Training updates in this campaign	0

These counts exclude ongoing validation, offline numerical checks and unfinished physical rollouts. Pending cells are not zeros. Same-start comparisons are paired observations; 40 starts do not create 40 independently trained models.

This report supplements the preserved earlier campaigns. Historical synthetic points, custom navigation, coverage-stress navigation and cube manipulation must not be pooled with the standard native episodes here.

What does the agent actually do?

Tasks 1 and 5 use native MuJoCo PointMaze navigation with real state transitions. A point agent selects two-dimensional control actions to reach the task goal while navigating the maze. This is navigation, separate from the earlier cube-manipulation study.

Task 1 navigates toward nominal goal (20,20). Task 5 starts near nominal (20,4) and navigates toward (0,0). Resets may vary within the task-defined start region. The standard evaluation cap is 1000 control decisions, separate from the earlier modified 128-step coverage test.

BC and IQL have different jobs. BC learns the offline action distribution. IQL learns Q(s,a) and V(s) from offline transitions including next states. This campaign freezes the BC actor from update 1000 and the IQL critic from update 50000; samplers propose actions but do not retrain Q or V.

The guidance score is the minimum of the two frozen target-Q heads. It is an estimate of future task return, not a certificate that a proposed action is safe or well supported by the offline data. “Unseen Q” here means evaluating this learned critic on actions outside or poorly covered by that data.

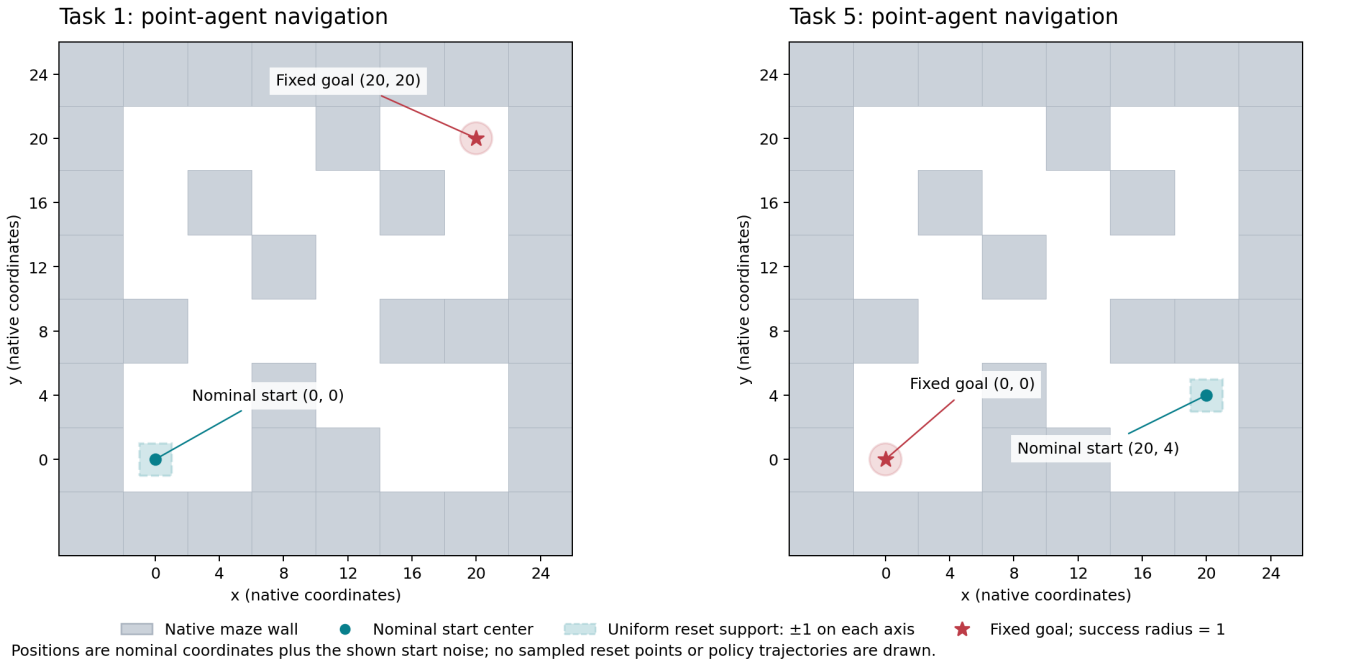
The native sparse reward is success minus one: -1 outside the radius-1 goal region, 0 on success. Success is checked before the action. The logged current position determines the relabeled reward and mask; IQL uses y=r+.99(1-success)V(actual next state), with V expectile .8. Larger Q predicts better discounted task return. The BC flow itself is ordinary behavior matching, with position input and no goal vector.

Seeds 11/23/37 identify independently trained actor/critic models. They are not task numbers or evaluation reset IDs. All three models remain in the report, including weak or failed ones.

Official maze geometry and task goals

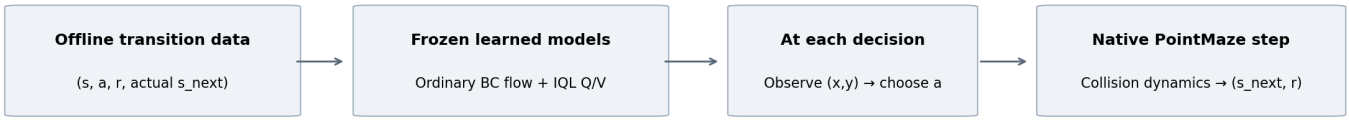
What the native experiments ask the agent to do

Navigate the point-agent around the actual walls to its fixed goal. Same official medium maze; different start and task reward.



Positions are nominal coordinates plus the shown start noise; no sampled reset points or policy trajectories are drawn.

Native task budget: up to 1,000 control decisions. Success is checked before the action; reward is −1 before success and 0 at success.



BC proposes actions; the fixed IQL-trained critic guides them. Q estimates discounted task reward and can be wrong; it is not an oracle.

These are sequential point-agent experiments with actual next states. The actor has 1,000 BC updates; the task-specific Q/V has 50,000 IQL updates.

Separate historical studies used a 128-decision TRAIN-anchor coverage stress or cube manipulation. Those resets, budgets and robot objectives differ.

Source-only drawing: official map/task literals and coordinate transform verified; no environment creation, reset, model call, physics or random draw.

Both tasks use this same official medium maze. Task1 starts around (0,0) and aims for (20,20); Task5 starts around (20,4) and aims for (0,0). Shaded start boxes show the native ±1 positional reset support, not sampled episodes. Red stars and radius-1 disks mark fixed goals. No policy route is drawn.

The figure comes from source-verified maze/task definitions and the exact coordinate transform. The model input is position (x,y); next states come from native collision dynamics. The flow time slices, D-Flow source iterations and physical control decisions are different time axes.

What κ, η, λ and averaging mean

Let F8(s,z) be the complete eight-slice frozen BC decoder including final action clipping. Raw S(s,a)=min(Q1,Q2). Averaged S is the mean over K nearby clipped actions of min(Q1,Q2); the minimum is taken before averaging.

Operation	Meaning in this campaign
κ (kappa)	Multiplier on Q-gradient guidance. Independently selected by task, trained model, method and fixed λ.
η (eta)	Step size for each D-Flow source update.
λ (lambda)	Coefficient of source-prior pull -λz. λ0 removes this pull; λ1 includes it.
I	Number of outer D-Flow source updates. Each update uses a full-flow endpoint.
σ / K	Nearby-action perturbation scale and total averaging samples. The antithetic bank stays fixed throughout one action generation.

Identity guidance: g = grad_a S(s,a), evaluated at a=F8(s,z), without multiplying by the endpoint Jacobian. Regular update: z ← z + η(κg - λz). Stochastic update adds sqrt(2ηT) ξ, with T=1. The executed action is F8(s,z_I).

Setting κ=0 removes Q guidance. It does not necessarily reproduce BC: λ1 still pulls z toward zero, and stochastic source noise still moves z. The unchanged BC flow remains a separate canonical control.

Source noise moves z. Q averaging changes the action score. Neither operation establishes correction of broad critic extrapolation error. Because the identity field is a surrogate, these runs do not imply exact Langevin sampling of Q(F8(z)). MH is outside this campaign’s scope.

Full-flow endpoints integrate all eight BC time slices. The excluded one-step D-Flow endpoint extrapolation is different from the numerical integrator used inside this complete flow. Original QGF still refreshes its own velocity correction and endpoint estimate at every flow time slice. QGF has no D-Flow η, λ or outer I setting.

Original QGF at each t=i/8: v_i=v(s,a_i,t); e_i=clip(a_i+(1-t)v_i); g_i=grad_a Q(s,a) evaluated at e_i; a_(i+1)=a_i+(v_i+κg_i)/8. It recomputes v_i, e_i and g_i at all eight slices, then applies its final action clipping. Its κ multiplies a velocity correction; D-Flow κ acts inside the repeated source update, so equal numerical κ values need not mean equal action changes.

Independent selection domain: κ=0,.02,.04,.06,.08,.1,.2,.3,.4,.6,.8,1,2,3,5,8,12,20,40,80; η=.005,.01,.03,.1; I=1,2,4,8,16,32,64,128,256; σ=.01,.03,.1,.2; K=4,16. Averaged-Q search uses 720 base settings plus at most 70 refinements per condition, followed by 8-episode physical validation. This hierarchy is not an exhaustive σ/K Cartesian sweep.

Frozen configurations: standard native results

Task 1: original A20 + fresh B20

Method	Seed 11	Seed 23	Seed 37
BC flow	0/40	Pending	Pending
QGF fixed .02	0/40	Pending	Pending
QGF fixed .04	0/40	Pending	Pending
QGF tuned	29/40	Pending	Pending
IPG	22/40	Pending	Pending
FK corrector	19/40	Pending	Pending
DF Regular, raw Q, λ0	28/40	Pending	Pending
DF Regular, raw Q, λ1	32/40	Pending	Pending
DF Stochastic, raw Q, λ0	24/40	Pending	Pending
DF Stochastic, raw Q, λ1	24/40	Pending	Pending

Task 5: new A20 + new B20

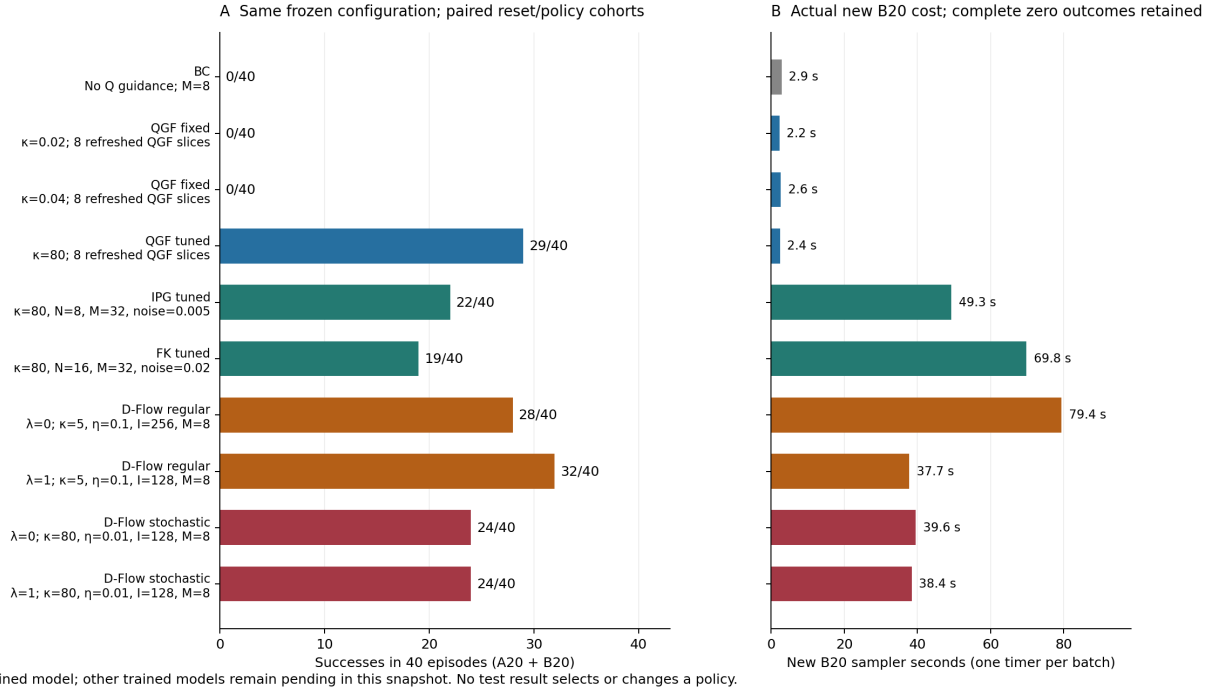
Method	Seed 11	Seed 23	Seed 37
BC flow	Pending	Pending	Pending
QGF fixed .02	Pending	Pending	Pending
QGF fixed .04	Pending	Pending	Pending
QGF tuned	Pending	Pending	Pending
IPG	Pending	Pending	Pending
FK corrector	Pending	Pending	Pending
DF Regular, raw Q, λ0	Pending	Pending	Pending
DF Regular, raw Q, λ1	Pending	Pending	Pending
DF Stochastic, raw Q, λ0	Pending	Pending	Pending
DF Stochastic, raw Q, λ1	Pending	Pending	Pending

Task1 may reference eligible original A20 and add fresh B20 at unchanged frozen settings. Task5 has its own separately declared new panel. All relevant three-model selections freeze before that branch's tests. Reported totals require both disjoint cohorts and an identical selected configuration.

First completed model: success and measured cost

Task 1, trained seed 11: frozen raw D-Flow versus QGF, IPG, FK and BC

Full-flow endpoint; identity Jacobian only for D-Flow. Each guided condition keeps its own validation-selected settings.



One independently trained model; other trained models remain pending in this snapshot. No test result selects or changes a policy.

Cost includes padded B20 execution and unequal early stopping, particle/flow counts and compilation effects; it is not a Q/NFE query count.

A20 is inherited evidence; B20 is new. Q-averaged identity methods use a separate fresh panel and are not represented by zero bars here.

This preserved figure shows Task1 training seed11 only: ten frozen policies on its original A20 plus fresh B20. The right panel times only newly executed B20 batches. It does not compare the new averaged-Q panel, which has separate starts and may still be pending.

Cost is measured once per batch. It includes unequal inference budgets and early stopping, and can include compilation effects. The three extra successes for regular λ1 relative to tuned QGF in this model do not establish a universal winner or isolate the effect of λ, since the selected iteration counts also differ.

New averaged-Q comparison: separately paired panel

Task 1: separate new fresh40 panel

Method	Seed 11	Seed 23	Seed 37
BC flow	Pending	Pending	Pending
QGF fixed .02	Pending	Pending	Pending
QGF fixed .04	Pending	Pending	Pending
QGF tuned	Pending	Pending	Pending
IPG	Pending	Pending	Pending
FK corrector	Pending	Pending	Pending
DF Regular, raw Q, λ0	Pending	Pending	Pending
DF Regular, raw Q, λ1	Pending	Pending	Pending
DF Stochastic, raw Q, λ0	Pending	Pending	Pending
DF Stochastic, raw Q, λ1	Pending	Pending	Pending
DF Regular, average Q, λ0	Pending	Pending	Pending
DF Regular, average Q, λ1	Pending	Pending	Pending
DF Stochastic, average Q, λ0	Pending	Pending	Pending
DF Stochastic, average Q, λ1	Pending	Pending	Pending

Task 5: separate new fresh40 panel

Method	Seed 11	Seed 23	Seed 37
BC flow	Pending	Pending	Pending
QGF fixed .02	Pending	Pending	Pending
QGF fixed .04	Pending	Pending	Pending
QGF tuned	Pending	Pending	Pending
IPG	Pending	Pending	Pending
FK corrector	Pending	Pending	Pending
DF Regular, raw Q, λ0	Pending	Pending	Pending
DF Regular, raw Q, λ1	Pending	Pending	Pending
DF Stochastic, raw Q, λ0	Pending	Pending	Pending
DF Stochastic, raw Q, λ1	Pending	Pending	Pending
DF Regular, average Q, λ0	Pending	Pending	Pending
DF Regular, average Q, λ1	Pending	Pending	Pending
DF Stochastic, average Q, λ0	Pending	Pending	Pending
DF Stochastic, average Q, λ1	Pending	Pending	Pending

Each fresh40 task branch requires all 33 guided choices across three trained models frozen before any heldout test. All 14 primary policies are reevaluated on that branch's same new starts. Its outcomes are not pooled with the different original A20+B20 extension panel. Display aliases do not create additional physical executions.

Selected settings (1-21 of 84)

Task / seed	Requested condition	κ	η / I	λ	σ / K	N / M	noise / ridge
1 / 11	BC flow	0	-	-	-	1 / 8	-
1 / 11	QGF fixed .02	0.02	-	-	-	1 / 8	-
1 / 11	QGF fixed .04	0.04	-	-	-	1 / 8	-
1 / 11	QGF tuned	80	-	-	-	1 / 8	-
1 / 11	IPG	80	-	-	-	8 / 32	0.005 / 0.1
1 / 11	FK corrector	80	-	-	-	16 / 32	0.02 / -
1 / 11	DF Regular, raw Q, λ0	5	0.1 / 256	0	-	1 / 8	-
1 / 11	DF Regular, raw Q, λ1	5	0.1 / 128	1	-	1 / 8	-
1 / 11	DF Stochastic, raw Q, λ0	80	0.01 / 128	0	-	1 / 8	-
1 / 11	DF Stochastic, raw Q, λ1	80	0.01 / 128	1	-	1 / 8	-
1 / 11	DF Regular, average Q, λ0	20	0.1 / 256	0	0.1 / 16	1 / 8	-
1 / 11	DF Regular, average Q, λ1 (pending)	-	-	-	-	-	-
1 / 11	DF Stochastic, average Q, λ0 (pending)	-	-	-	-	-	-
1 / 11	DF Stochastic, average Q, λ1 (pending)	-	-	-	-	-	-
1 / 23	BC flow	0	-	-	-	1 / 8	-
1 / 23	QGF fixed .02	0.02	-	-	-	1 / 8	-
1 / 23	QGF fixed .04	0.04	-	-	-	1 / 8	-
1 / 23	QGF tuned	80	-	-	-	1 / 8	-
1 / 23	IPG	40	-	-	-	16 / 32	0.02 / 0.01
1 / 23	FK corrector	80	-	-	-	16 / 32	0.02 / -
1 / 23	DF Regular, raw Q, λ0	40	0.1 / 256	0	-	1 / 8	-

N=particles; M=flow time slices. A dash means unused or still pending, not zero. Stochastic D-Flow uses T=1; particle noise is a separate setting. FK does not use IPG ridge. Complete configuration dictionaries and selection states are in SELECTED_SETTINGS.json. Individually selected new-Qavg settings may still await the all-three-model test barrier.

Selected settings (22-42 of 84)

Task / seed	Requested condition	κ	η / I	λ	σ / K	N / M	noise / ridge
1 / 23	DF Regular, raw Q, λ1	80	0.03 / 256	1	-	1 / 8	-
1 / 23	DF Stochastic, raw Q, λ0	80	0.03 / 256	0	-	1 / 8	-
1 / 23	DF Stochastic, raw Q, λ1	80	0.03 / 128	1	-	1 / 8	-
1 / 23	DF Regular, average Q, λ0	40	0.1 / 256	0	0.2 / 16	1 / 8	-
1 / 23	DF Regular, average Q, λ1	80	0.03 / 256	1	0.2 / 4	1 / 8	-
1 / 23	DF Stochastic, average Q, λ0 (pending)	-	-	-	-	-	-
1 / 23	DF Stochastic, average Q, λ1 (pending)	-	-	-	-	-	-
1 / 37	BC flow	0	-	-	-	1 / 8	-
1 / 37	QGF fixed .02	0.02	-	-	-	1 / 8	-
1 / 37	QGF fixed .04	0.04	-	-	-	1 / 8	-
1 / 37	QGF tuned	40	-	-	-	1 / 8	-
1 / 37	IPG	40	-	-	-	16 / 8	0.02 / 0.01
1 / 37	FK corrector	80	-	-	-	16 / 32	0.02 / -
1 / 37	DF Regular, raw Q, λ0	5	0.1 / 256	0	-	1 / 8	-
1 / 37	DF Regular, raw Q, λ1	80	0.03 / 256	1	-	1 / 8	-
1 / 37	DF Stochastic, raw Q, λ0	80	0.1 / 256	0	-	1 / 8	-
1 / 37	DF Stochastic, raw Q, λ1	80	0.03 / 128	1	-	1 / 8	-
1 / 37	DF Regular, average Q, λ0 (pending)	-	-	-	-	-	-
1 / 37	DF Regular, average Q, λ1 (pending)	-	-	-	-	-	-
1 / 37	DF Stochastic, average Q, λ0 (pending)	-	-	-	-	-	-
1 / 37	DF Stochastic, average Q, λ1 (pending)	-	-	-	-	-	-

N=particles; M=flow time slices. A dash means unused or still pending, not zero. Stochastic D-Flow uses T=1; particle noise is a separate setting. FK does not use IPG ridge. Complete configuration dictionaries and selection states are in SELECTED_SETTINGS.json. Individually selected new-Qavg settings may still await the all-three-model test barrier.

Selected settings (43-63 of 84)

Task / seed	Requested condition	κ	η / I	λ	σ / K	N / M	noise / ridge
5 / 11	BC flow	0	-	-	-	1 / 8	-
5 / 11	QGF fixed .02	0.02	-	-	-	1 / 8	-
5 / 11	QGF fixed .04	0.04	-	-	-	1 / 8	-
5 / 11	QGF tuned	20	-	-	-	1 / 8	-
5 / 11	IPG	20	-	-	-	16 / 32	0.005 / 0.01
5 / 11	FK corrector	80	-	-	-	8 / 32	0.02 / -
5 / 11	DF Regular, raw Q, λ0	1	0.1 / 256	0	-	1 / 8	-
5 / 11	DF Regular, raw Q, λ1	5	0.1 / 128	1	-	1 / 8	-
5 / 11	DF Stochastic, raw Q, λ0	80	0.1 / 256	0	-	1 / 8	-
5 / 11	DF Stochastic, raw Q, λ1	5	0.03 / 128	1	-	1 / 8	-
5 / 11	DF Regular, average Q, λ0 (pending)	-	-	-	-	-	-
5 / 11	DF Regular, average Q, λ1 (pending)	-	-	-	-	-	-
5 / 11	DF Stochastic, average Q, λ0 (pending)	-	-	-	-	-	-
5 / 11	DF Stochastic, average Q, λ1 (pending)	-	-	-	-	-	-
5 / 23	BC flow	0	-	-	-	1 / 8	-
5 / 23	QGF fixed .02	0.02	-	-	-	1 / 8	-
5 / 23	QGF fixed .04	0.04	-	-	-	1 / 8	-
5 / 23	QGF tuned	40	-	-	-	1 / 8	-
5 / 23	IPG	5	-	-	-	8 / 8	0 / 0.01
5 / 23	FK corrector	5	-	-	-	16 / 32	0 / -
5 / 23	DF Regular, raw Q, λ0	5	0.1 / 128	0	-	1 / 8	-

N=particles; M=flow time slices. A dash means unused or still pending, not zero. Stochastic D-Flow uses T=1; particle noise is a separate setting. FK does not use IPG ridge. Complete configuration dictionaries and selection states are in SELECTED_SETTINGS.json. Individually selected new-Qavg settings may still await the all-three-model test barrier.

Selected settings (64-84 of 84)

Task / seed	Requested condition	κ	η / l	λ	σ / K	N / M	noise / ridge
5 / 23	DF Regular, raw Q, λ1	5	0.1 / 128	1	-	1 / 8	-
5 / 23	DF Stochastic, raw Q, λ0	5	0.1 / 128	0	-	1 / 8	-
5 / 23	DF Stochastic, raw Q, λ1	40	0.01 / 2	1	-	1 / 8	-
5 / 23	DF Regular, average Q, λ0 (pending)	-	-	-	-	-	-
5 / 23	DF Regular, average Q, λ1 (pending)	-	-	-	-	-	-
5 / 23	DF Stochastic, average Q, λ0 (pending)	-	-	-	-	-	-
5 / 23	DF Stochastic, average Q, λ1 (pending)	-	-	-	-	-	-
5 / 37	BC flow	0	-	-	-	1 / 8	-
5 / 37	QGF fixed .02	0.02	-	-	-	1 / 8	-
5 / 37	QGF fixed .04	0.04	-	-	-	1 / 8	-
5 / 37	QGF tuned	40	-	-	-	1 / 8	-
5 / 37	IPG	8	-	-	-	8 / 32	0 / 0.01
5 / 37	FK corrector	40	-	-	-	16 / 8	0.02 / -
5 / 37	DF Regular, raw Q, λ0	5	0.1 / 128	0	-	1 / 8	-
5 / 37	DF Regular, raw Q, λ1	80	0.1 / 16	1	-	1 / 8	-
5 / 37	DF Stochastic, raw Q, λ0	80	0.1 / 8	0	-	1 / 8	-
5 / 37	DF Stochastic, raw Q, λ1	80	0.1 / 8	1	-	1 / 8	-
5 / 37	DF Regular, average Q, λ0 (pending)	-	-	-	-	-	-
5 / 37	DF Regular, average Q, λ1 (pending)	-	-	-	-	-	-
5 / 37	DF Stochastic, average Q, λ0 (pending)	-	-	-	-	-	-
5 / 37	DF Stochastic, average Q, λ1 (pending)	-	-	-	-	-	-

N=particles; M=flow time slices. A dash means unused or still pending, not zero. Stochastic D-Flow uses T=1; particle noise is a separate setting. FK does not use IPG ridge. Complete configuration dictionaries and selection states are in SELECTED_SETTINGS.json. Individually selected new-Qavg settings may still await the all-three-model test barrier.

Failures, limits and numerical checks

All six model math/profile gates completed. The preserved raw float32 finite-difference checks include 6 failures out of 774 raw rows. Separately declared reference criteria passed: 678 non-FD checks and 96 double-reference comparisons, with 0 qualified failures. No tolerance, checkpoint, production precision or numerical-source change was made to erase the raw failures.

Engineering profile: 84 settings, 168 offline sampler calls and 3360 action rows. These are not physical episodes. The largest observed steady B20 call was 0.371 seconds; this is not a full-episode runtime guarantee.

A source-description error was found and corrected before fresh40 release: actual reset IDs, not row indices 0-19, enter the source-key folding. Disjoint A/B reset IDs create 40 distinct episode-key contexts under a shared policy root. The old wording is preserved; numerical code and cohorts were unchanged.

A repeated canonical BC validation control can consume real compute while reproducing already observed trajectories. Such repeats count as executions/cost but not new independent trajectory evidence. Raw completed-record counters after a resume may include an earlier partial prefix; report unique completed evidence or subtract separately preserved prefix work before claiming incremental attempt cost.

Q averaging may suppress a narrow false peak or a genuine narrow optimum. Noise may help movement within the actor’s image or amplify a mistaken score. Native success alone does not identify which mechanism occurred. Learned-V Bellman residuals measure self-consistency, not ground-truth Q error. Saved traces do not contain every latent, perturbation bank or counterfactual Q value.

Small validation panels and hierarchical search can miss useful settings or select noisy winners. Unequal inference iterations, particle/flow counts, early stopping and compilation confound runtime comparisons. One trained model cannot establish a universal method winner. All completed zeros remain in the result tables; pending and interrupted cells remain explicit.

No new full Task2 comparison was released in this campaign. Tasks1 and5 retain priority under the declared local-compute budget; absent Task2 cells are not successful or failed runs.

Reproducibility and preserved campaigns

The evidence JSON binds complete CSV/metadata/trace triples with SHA256, exact reset IDs, policy roots, model inputs and selected configurations. New physical evidence and inherited references are counted separately. Selection uses validation only; heldout outcomes never retune or choose a family configuration.

Native model construction combines the frozen BC policy with the frozen goal-specific IQL critic. The two-head target minimum is used for guidance. Shared model hashes and full-agent tree hashes are checked by released workers before/after execution. Local CPU only: at most two scientific workers, bounded 7200-second attempts, explicit reviewed checkpoint resumes, no automatic failure retries.

Task / seed	BC checkpoint SHA prefix	IQL checkpoint SHA prefix
1 / 11	45496ed77dd755a7	9bb2d24f10135780
1 / 23	ce24a7f3c86bbfaa	e5e09273f762357a
1 / 37	1a01ebdb0ed1d8a9	995c4299ccabef98
5 / 11	45496ed77dd755a7	fc61e43f2794c8c4
5 / 23	ce24a7f3c86bbfaa	d07c86d71ed81012
5 / 37	1a01ebdb0ed1d8a9	d6ec49560a722a9f

Verified common configuration: H=1, flow slices=8, two Q heads with minimum aggregation; discount=0.99, expectile=0.8, target τ=0.005, batch=256, learning rates=0.0003. Actor hidden layers=[64, 64]; Q/V hidden layers=[128, 128]. Full checkpoint paths, configuration dictionaries and SHA256 values are included in the bound manifest/reference bundle.

Core numerical source hashes

work/overnight_research_20261105/samplers.py
a615c61d284cb256e1291a5ae71cc719b03614a9218f84a49e86dfa6a80105e9

work/overnight_research_20261106_night/identity_math/kernel.py
0f8d7a4519ba95cc3e8aafcfc3996e381408c9dbc44ac555cb0ccae3cab79d56

work/overnight_research_20261106_continuation/exploratory_identity_qavg_v2/kernel.py
7d7d10799f5f3b81aad15668568c477aaf95bffe35949c47b701ec753ba01ddc

Historical reports and archives remain separate. The earlier all-experiments report includes synthetic one-point diagnostics, custom sequential learned-BC/IQL navigation, modified FULL/HOLE coverage studies and MuJoCo cube manipulation. FULL/HOLE labels concern older offline-data coverage arms; they do not mean full-flow versus Euler endpoints here.

Final archive verification: pending the final morning integrity receipt; this preview is not final delivery.

A next discriminating experiment should be preregistered after this study and use new data: match inference budgets, repeat trained-model seeds, and separate local narrow-peak errors from broad critic errors using a controlled reference. No such new experiment or retuning is performed from these heldout results.